

| Question | Marking Guidance                                                                                                                                                                                                                                                                     | Mark  | Comments                                                                                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 06.1     | 1. (Usually) Type II produce insulin;<br>2. Cells/receptors less sensitive/responsive (to insulin)<br>OR<br>Faulty (insulin) receptors;<br>3. (Treated/controlled by) diet/exercise;                                                                                                 | 2 max | 2. Accept: cells/receptors do not respond.<br>2. Accept: 'fewer receptors'<br>3. Accept: (Treated/controlled by) weight loss/medication/drugs.<br>3. Ignore: diabetes is caused by diet/exercise. |
| 06.2     | Auto-marked – tick in box 4                                                                                                                                                                                                                                                          | 1     |                                                                                                                                                                                                   |
| 06.3     | 1. Attach to gene/DNA/promoter region;<br>2. Stimulate/inhibit transcription/RNA polymerase;                                                                                                                                                                                         | 2     | Note: Genes being expressed/inhibited or switched on/off is not enough on its own.                                                                                                                |
| 06.4     | 1. (Effective as) group A/with iPS/treated lower than group B/with diabetes;<br>2. (Effective as) group A similar to group C/without diabetes;<br>3. (Investigation) done on mice not humans;<br>4. Only shows results for 12 weeks/short-time period / long-term effects not known; | 4     | Ignore: Only one study / not repeated / sample size.<br>2. Accept: 'healthy' or 'normal' or control for group C.                                                                                  |

